

Story 2: Can the FED Control Inflation and Maintain Full Employment (Last 25 Years)

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1 Purpose and Research Questions

The Federal Reserve's dual mandate from Congress is to:

1. Maintain **stable prices** (control inflation)
2. Maintain **maximum employment** (low unemployment)

These objectives can conflict, especially around recessions and supply shocks.

Main Question:

Has the FED been able to fulfill the mandate given to it by Congress over the last 25 years?

This analysis is **observational** and focuses on time ordering and consistency with known policy lags. It does **not** claim causal proof.

2 Story Development Focus

Focus Point: *Visual selection framework + slide messaging technique*

- Use a small set of repeatable, consistent visuals (simplicity + uniformity)
- Use clear benchmark lines and shaded “episodes” to emphasize “good vs bad” periods (saliency)
- Use separate panels rather than dual axes to avoid distortion (fidelity + efficacy)
- Use a lag-aware view to respect the timing relationship (Fed reacts to inflation; unemployment responds later)

3 Data Sources (API)

We pull monthly data for the last 25 years via API:

- **CPI-U (All items, SA)** — BLS series: CUSR0000SA0
- **Unemployment Rate** — BLS series: LNS14000000
- **Effective Federal Funds Rate** — FRED series: FEDFUNDS

API keys (recommended):

- BLS v2 registration key: set BLS_KEY in `.Renvi`
- FRED API key: set FRED_KEY in `.Renvi`

4 Theory

The core policy theory is that the Fed can move one tool, the policy interest rate, to influence both sides of its mandate over time:

- Higher interest rates reduce borrowing, spending, and pricing pressure, helping inflation move down.
- Lower interest rates support borrowing and demand, helping employment recover when labor markets weaken.

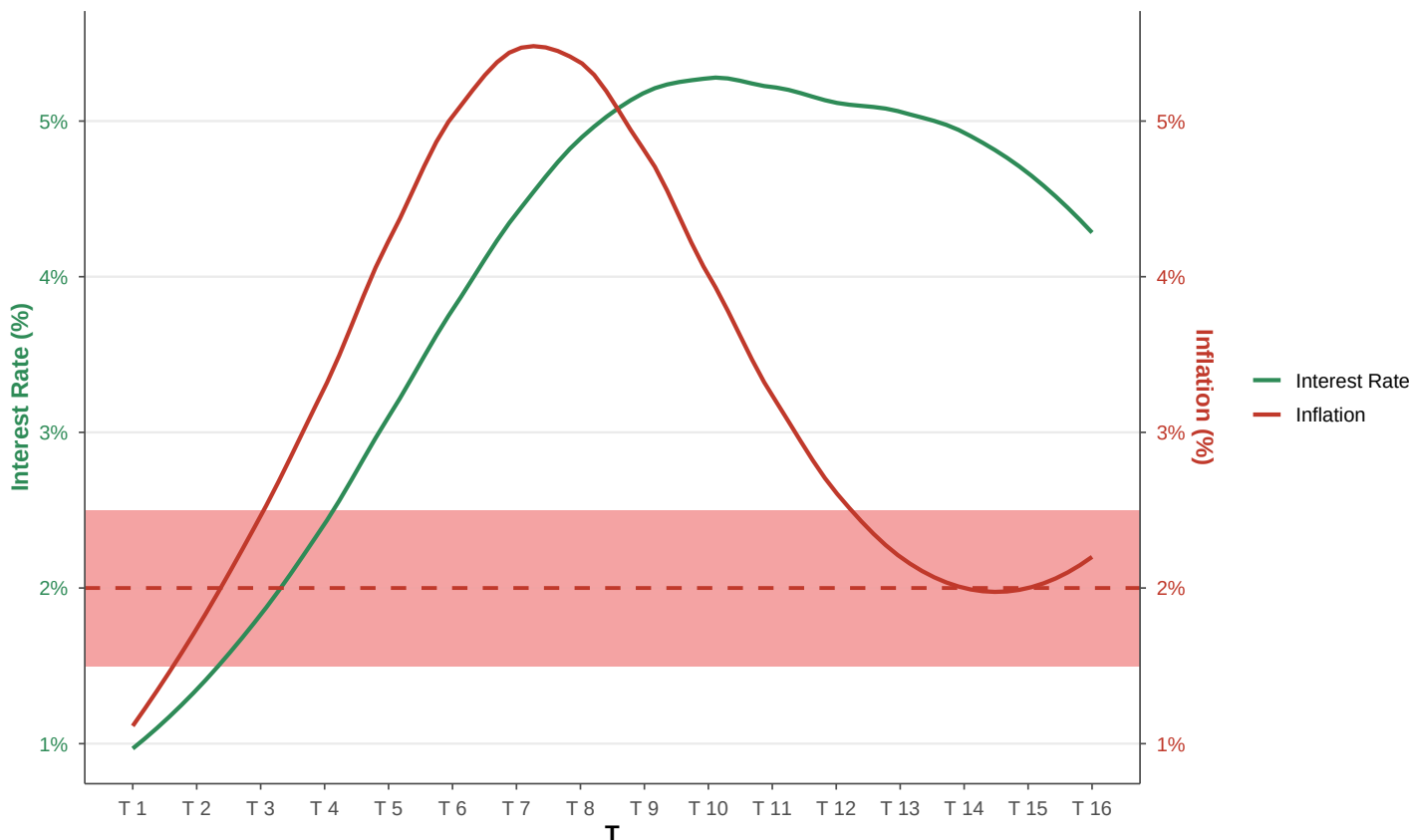
In practice, this is not instant. Inflation and unemployment usually respond with lags, and shocks can temporarily break the tradeoff.

Band legend for theory visuals: - Light red band = inflation benchmark margin-of-error around the dashed inflation target line. - Light blue band = unemployment benchmark margin-of-error around the dashed unemployment benchmark line.

4.1 Theory Slide 1: Interest Rates and Inflation

Theory (Best Case) Inverse: Interest Rate Hikes Lead Inflation Lower With a Lag

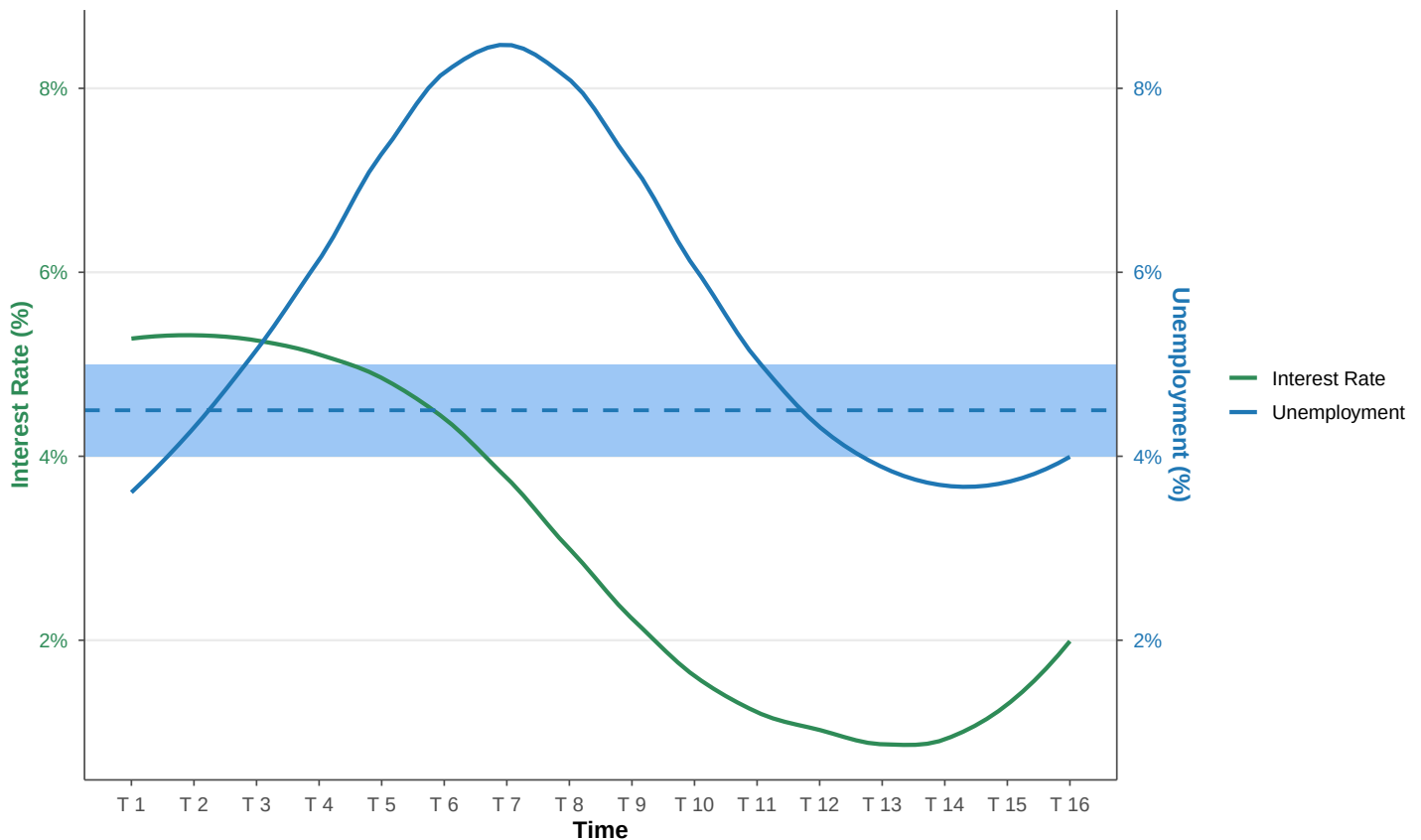
Light red band: inflation benchmark margin-of-error around the dashed target line



4.2 Theory Slide 2: Interest Rates and Unemployment

Theory (Best Case) Direct: Unemployment Rises First, Then Rate Cuts Follow and Unemployment Falls

Light blue band: unemployment benchmark margin-of-error around the dashed benchmark line



5 API Helpers

6 Episodes to Highlight (Good vs Bad Performance)

These shaded windows act like “chapters” in the story.

7 Pull and Prepare Data (Last 25 Years)

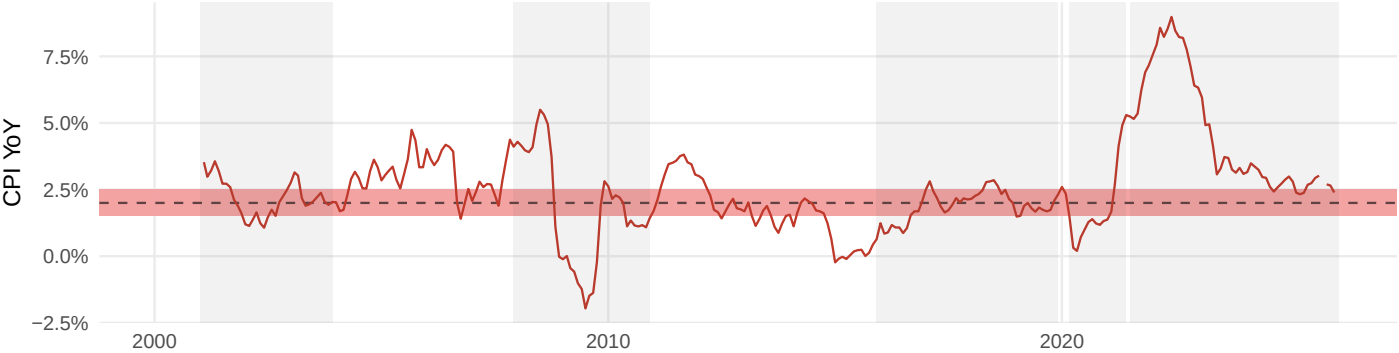
8 Visual Theme (Uniformity + Simplicity)

9 Visualization 1 — Mandate Scorecard Timeline (No Dual Axis)

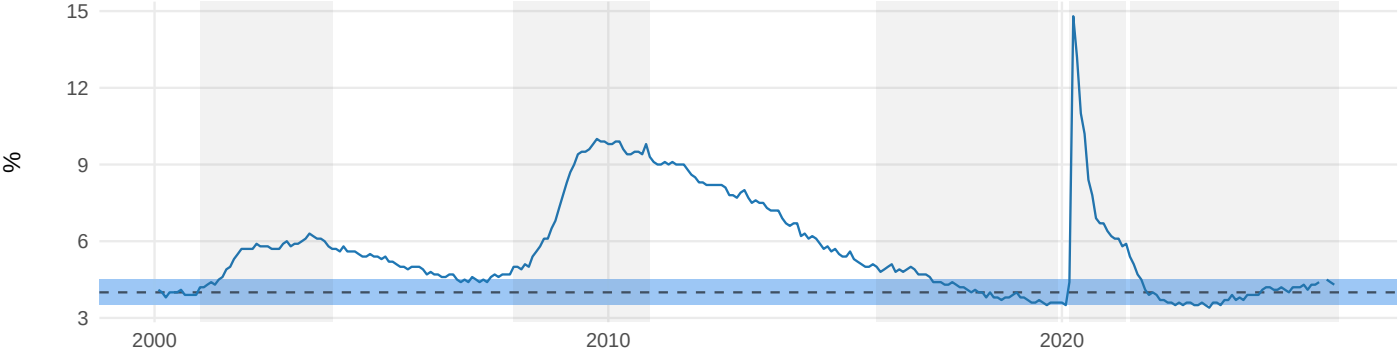
This stacks three aligned time series (inflation, unemployment, policy rate).

Fed mandate scorecard (25 years): inflation, unemployment, and the policy lever

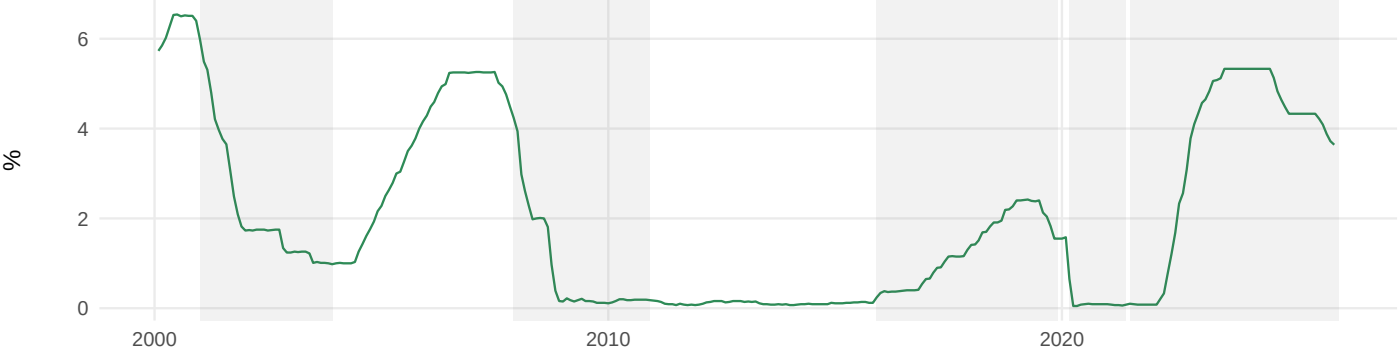
Inflation (CPI YoY) vs Fed mandate benchmark



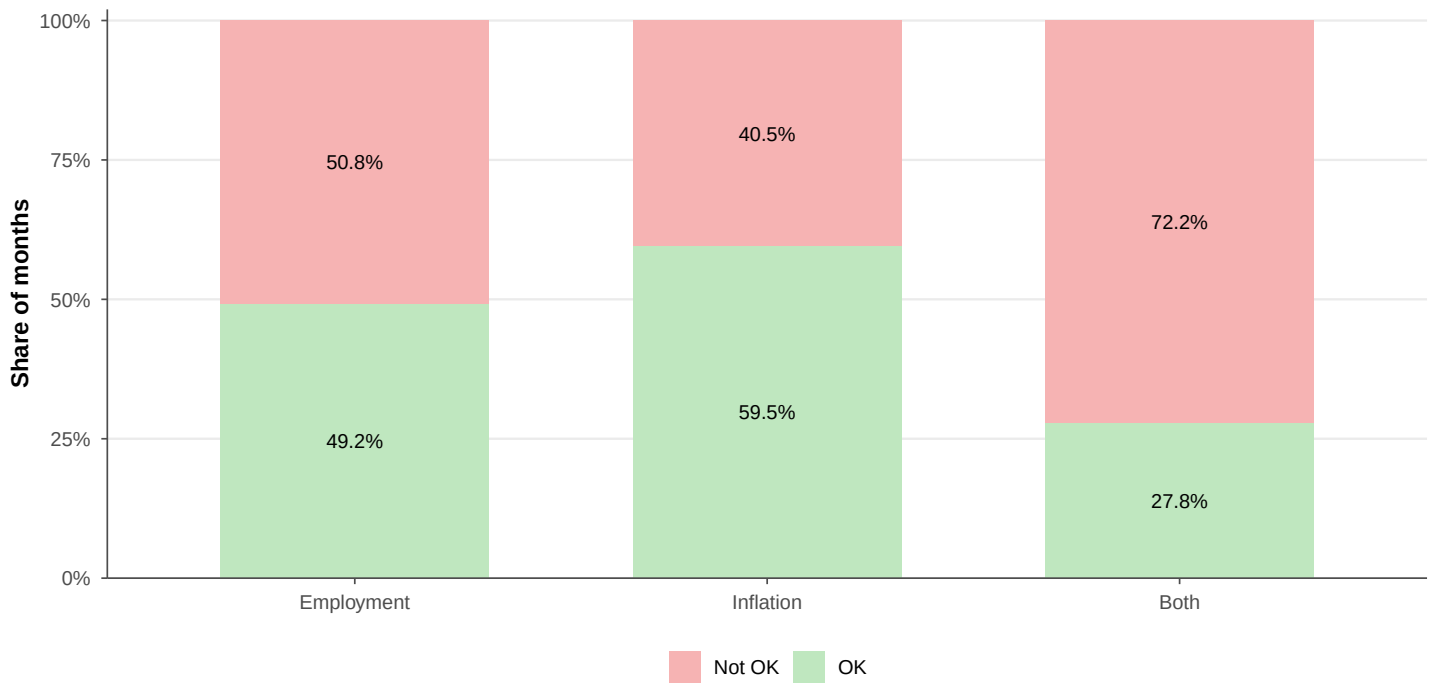
Unemployment rate



Policy tool: Effective Fed Funds Rate



26-year mandate benchmark share (monthly): Employment, Inflation, and Both



10 Period deep-dives (Theory-style charts)

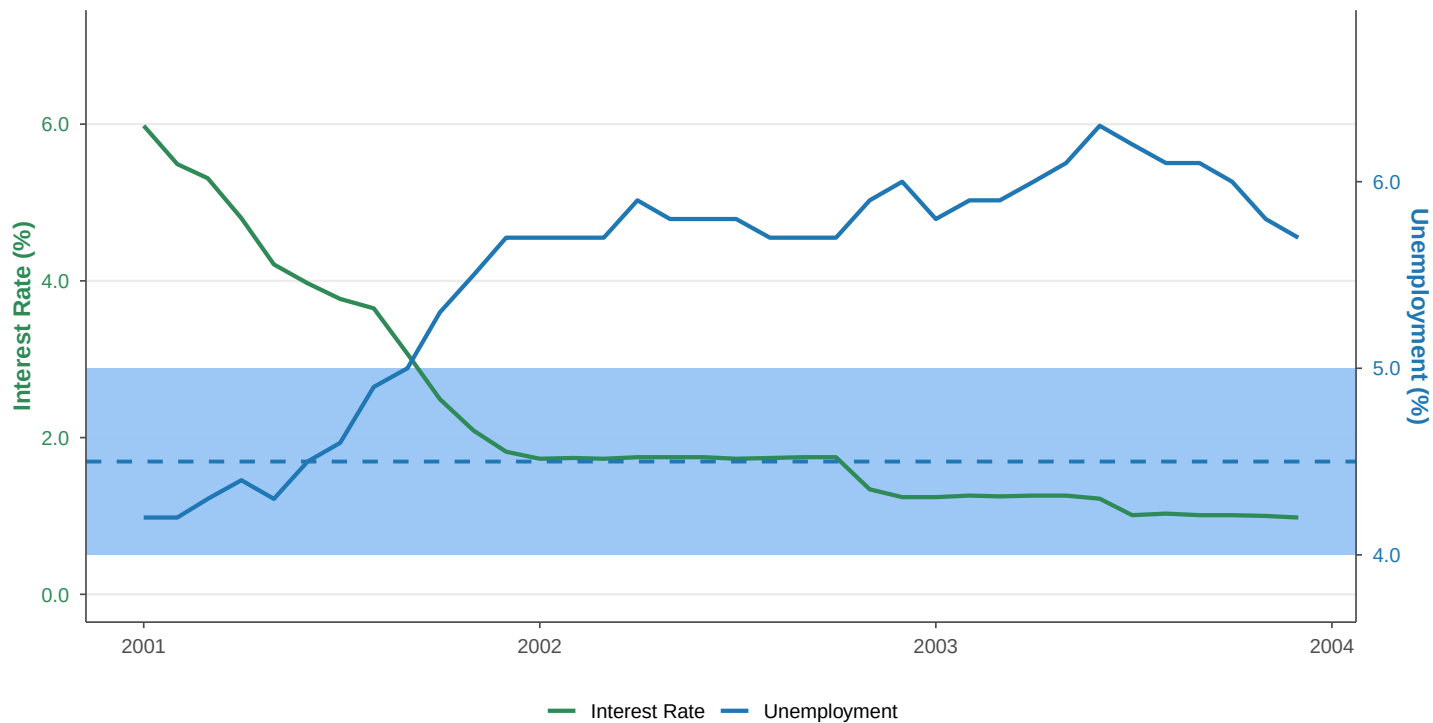
For each period of interest, we reproduce **two theory-style, dual-axis line charts**:

- 1) **Fed Funds vs Unemployment** (dashed = unemployment, right axis)
- 2) **Fed Funds vs Inflation (CPI YoY)** (dashed = inflation, right axis)

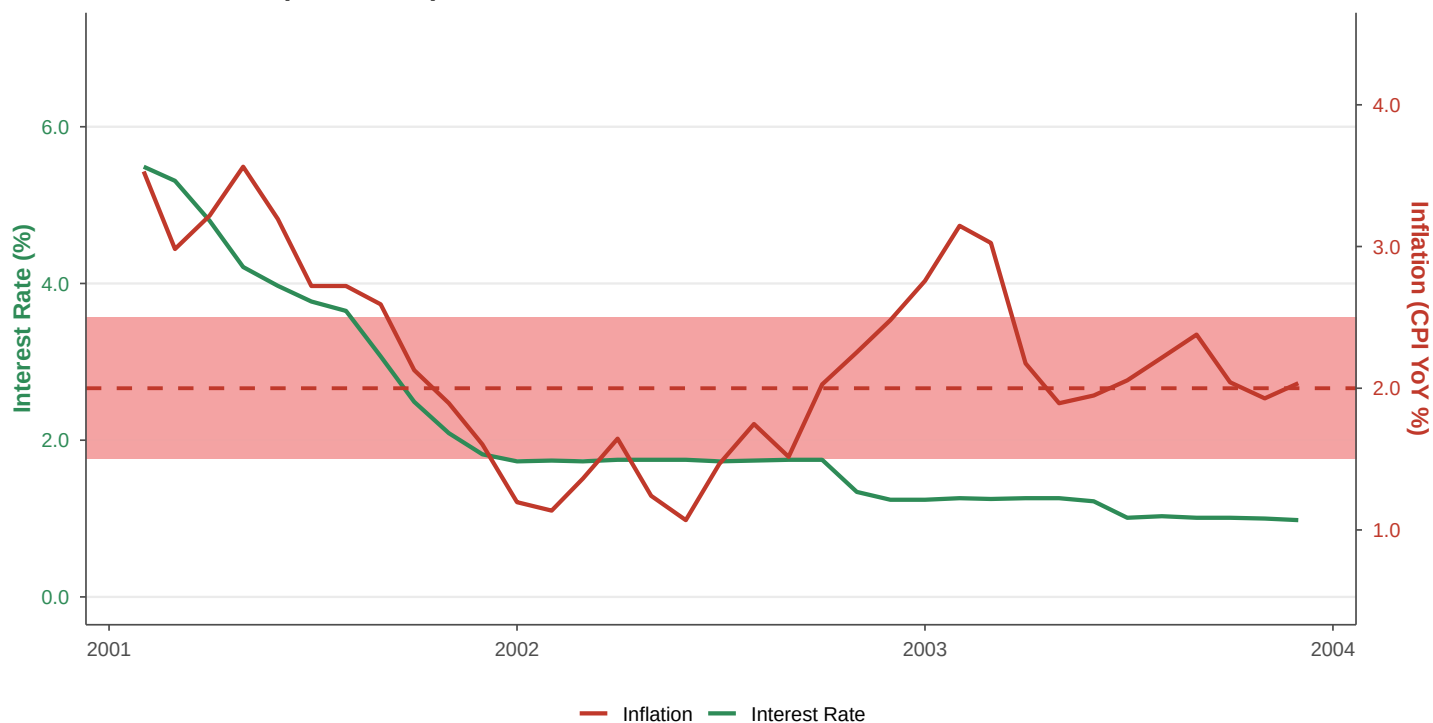
Each chart **rescales the right axis per-period** so the two lines are visually comparable and the relationship is clear.

10.1 Dot-com bust (2000-2003)

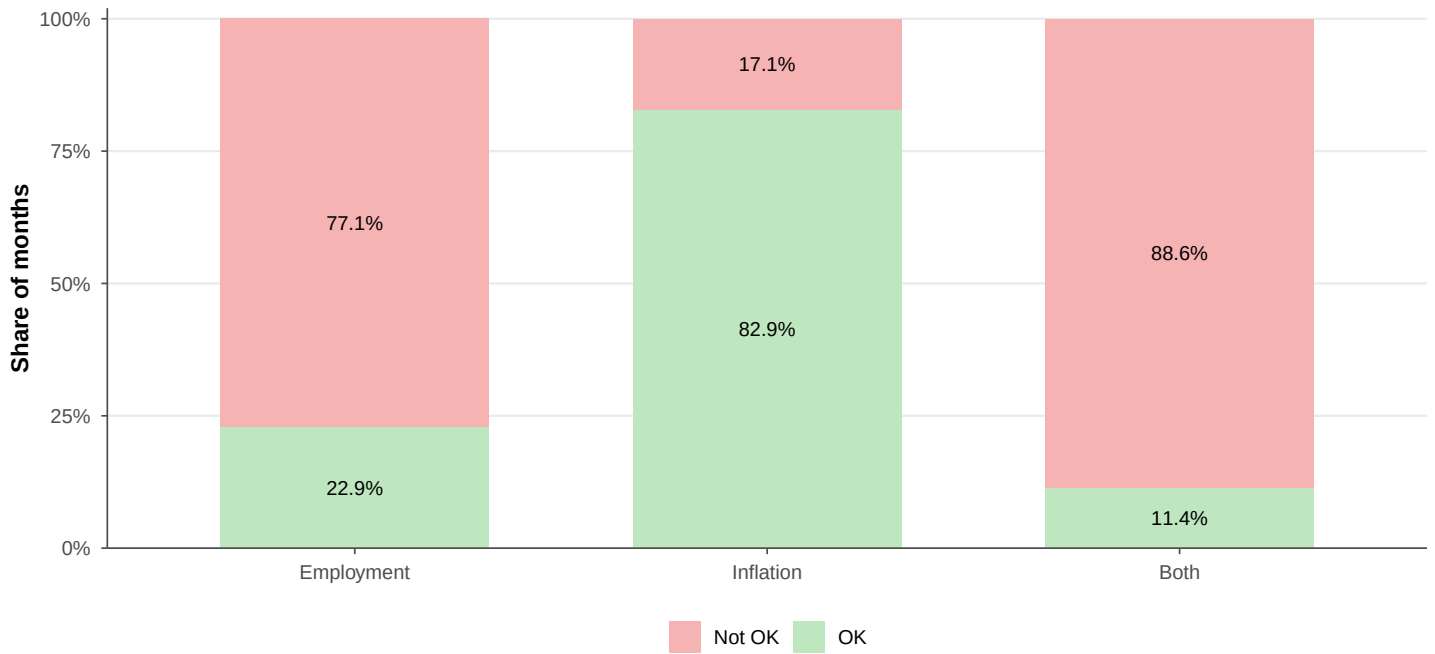
Dot-com bust (2000–2003): Fed Funds vs Unemployment



Dot-com bust (2001–2003): Fed Funds vs Inflation

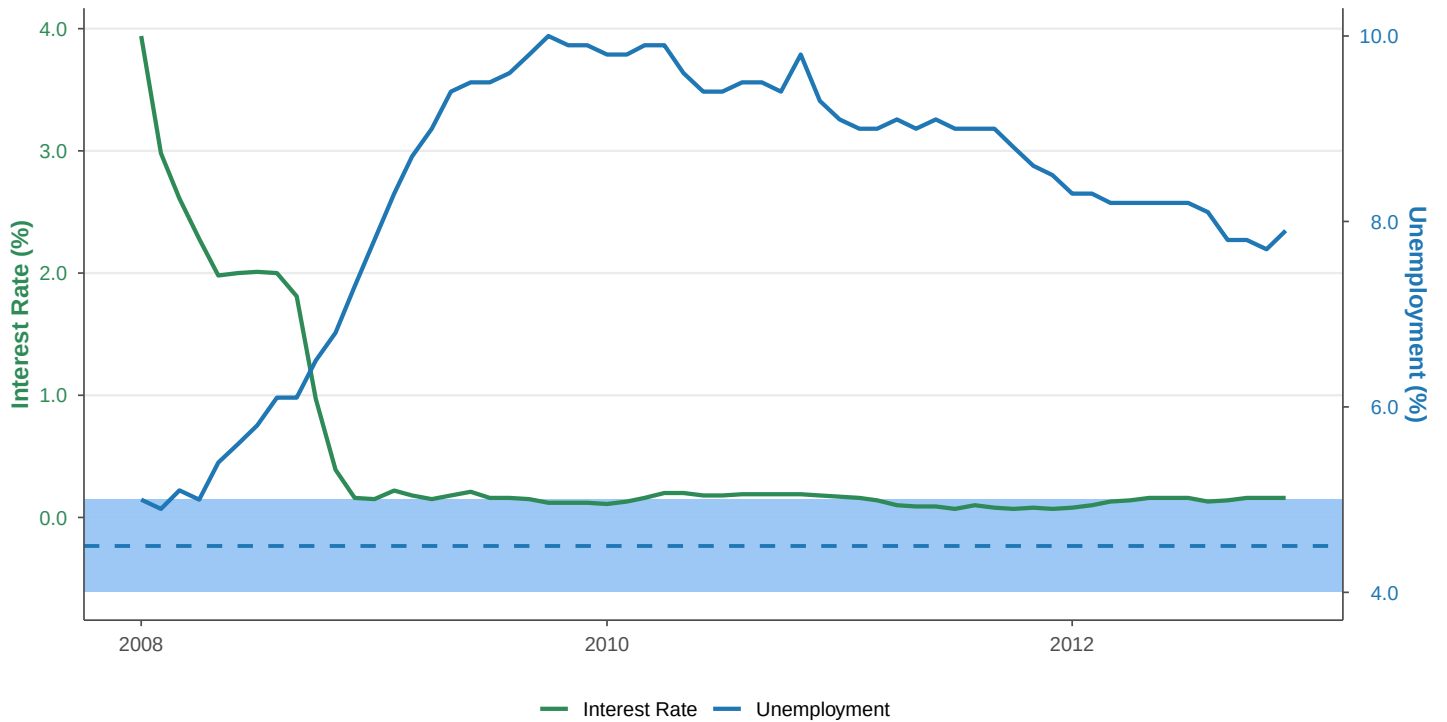


Dot-com bust (2000–2003): Percentage of months meeting benchmarks

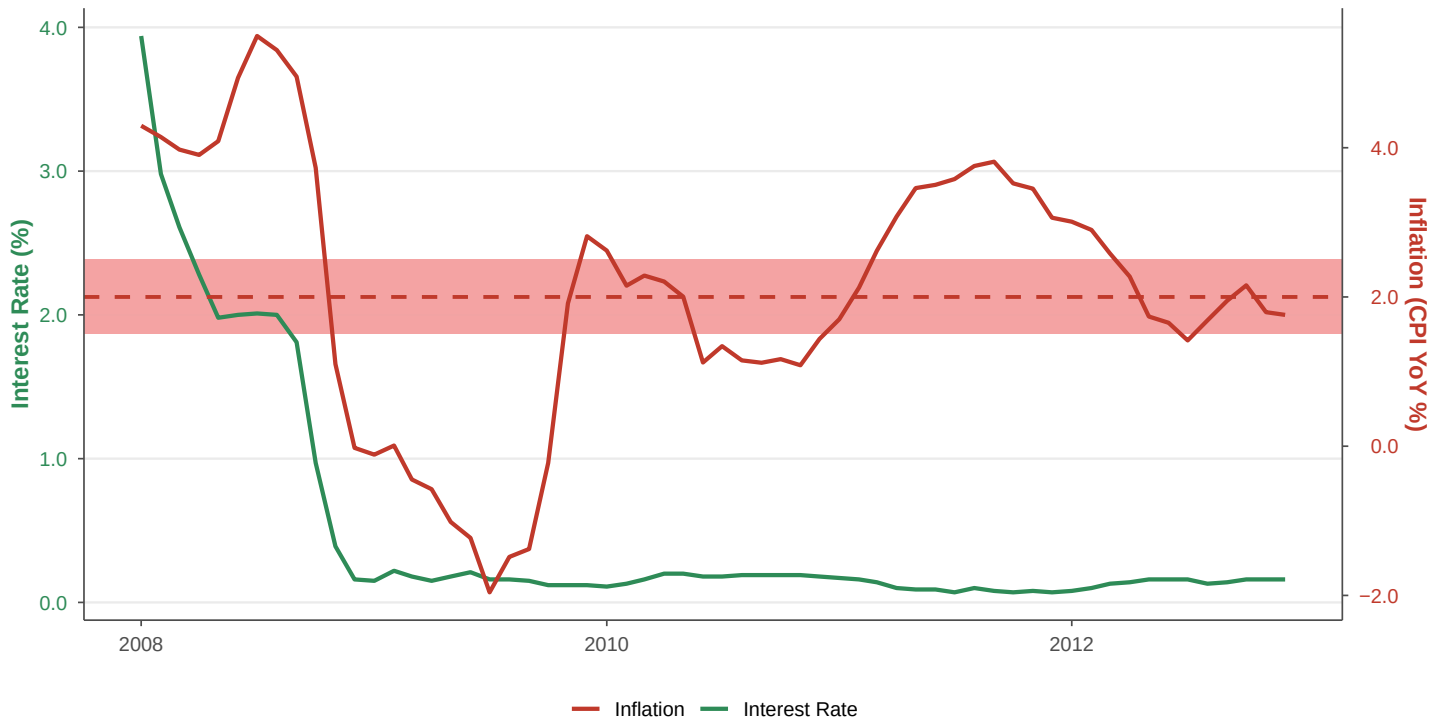


10.2 Financial crisis + aftermath (2008-2012)

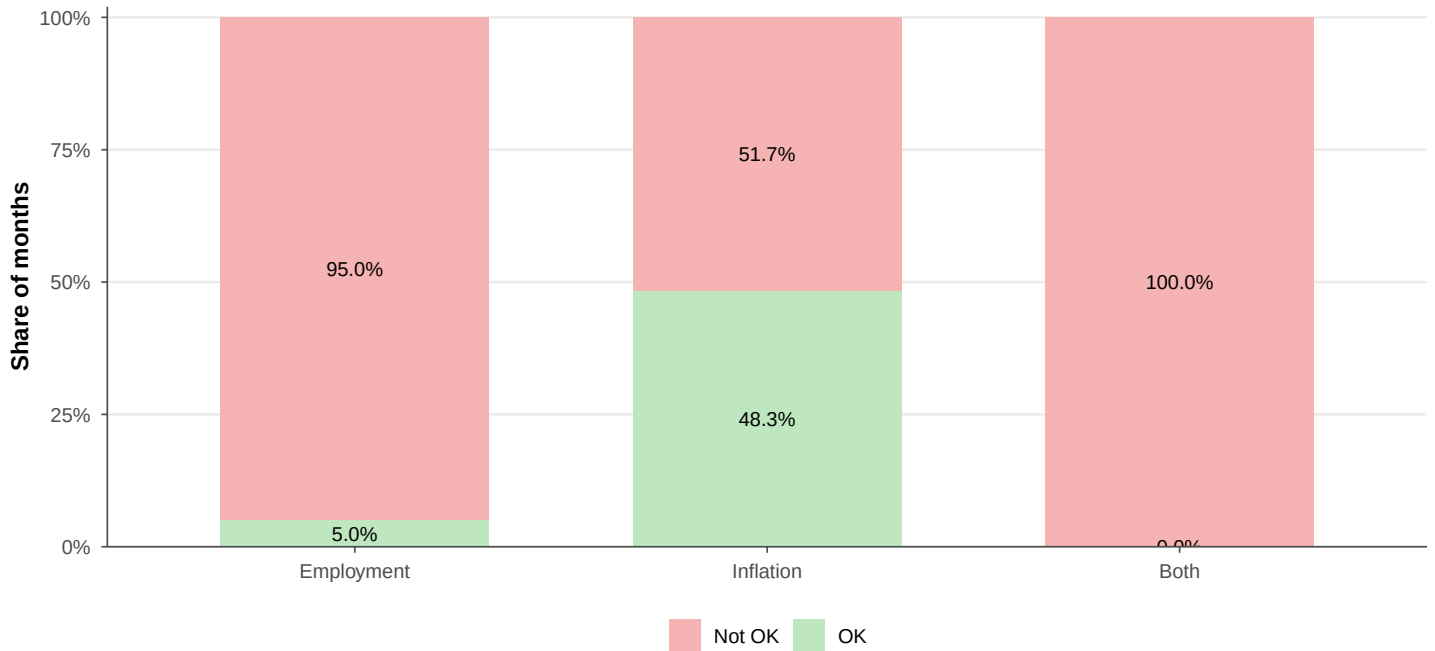
Financial crisis + aftermath (2008–2012): Fed Funds vs Unemployment



Financial crisis + aftermath (2008–2012): Fed Funds vs Inflation (CPI YoY)

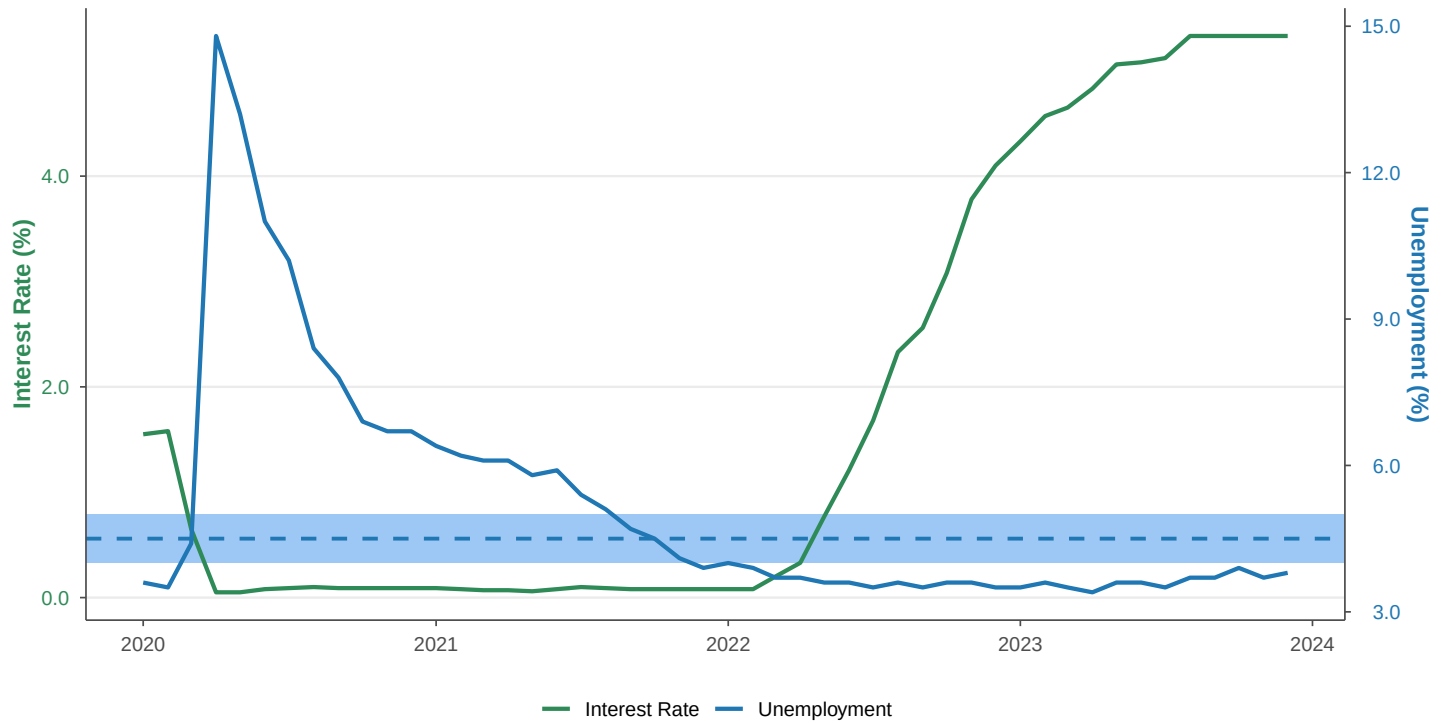


Financial crisis + aftermath (2008–2012): Percentage of months meeting benchmarks

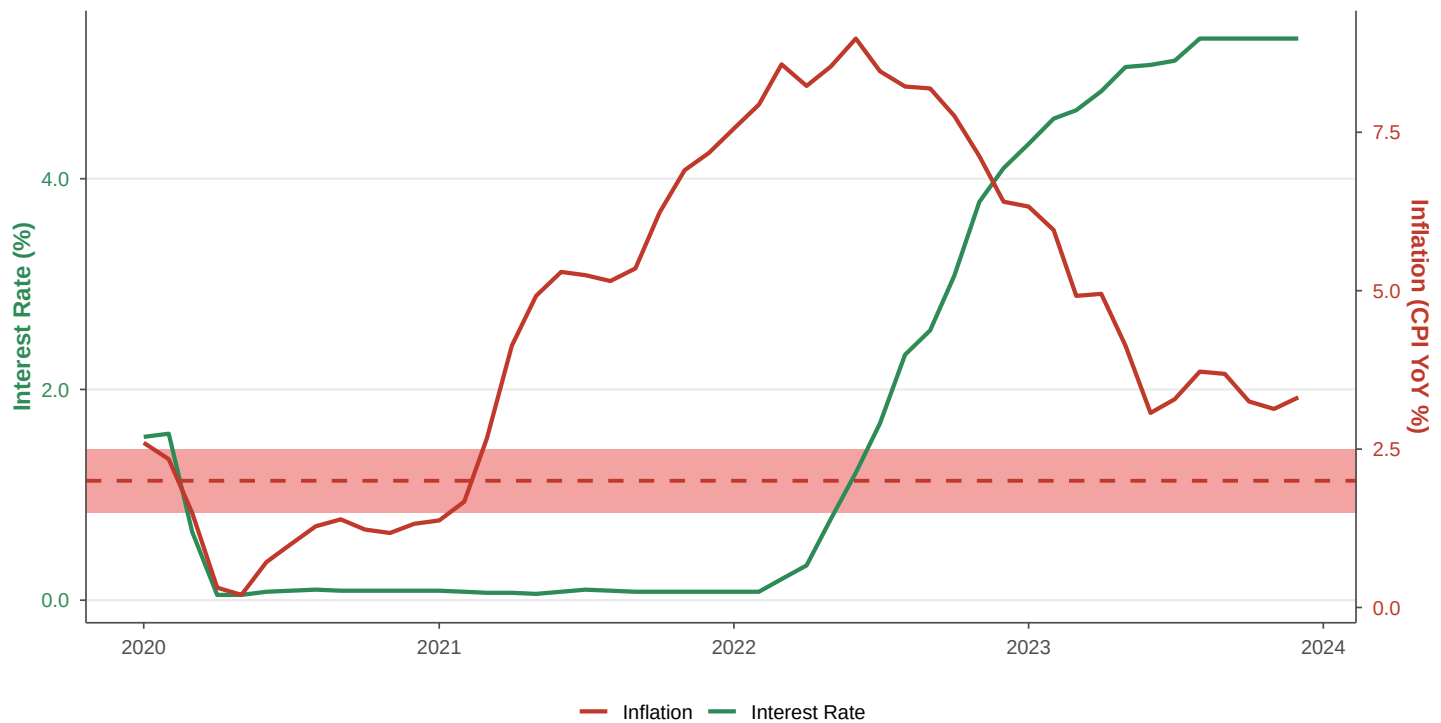


10.3 COVID era (2020-2023)

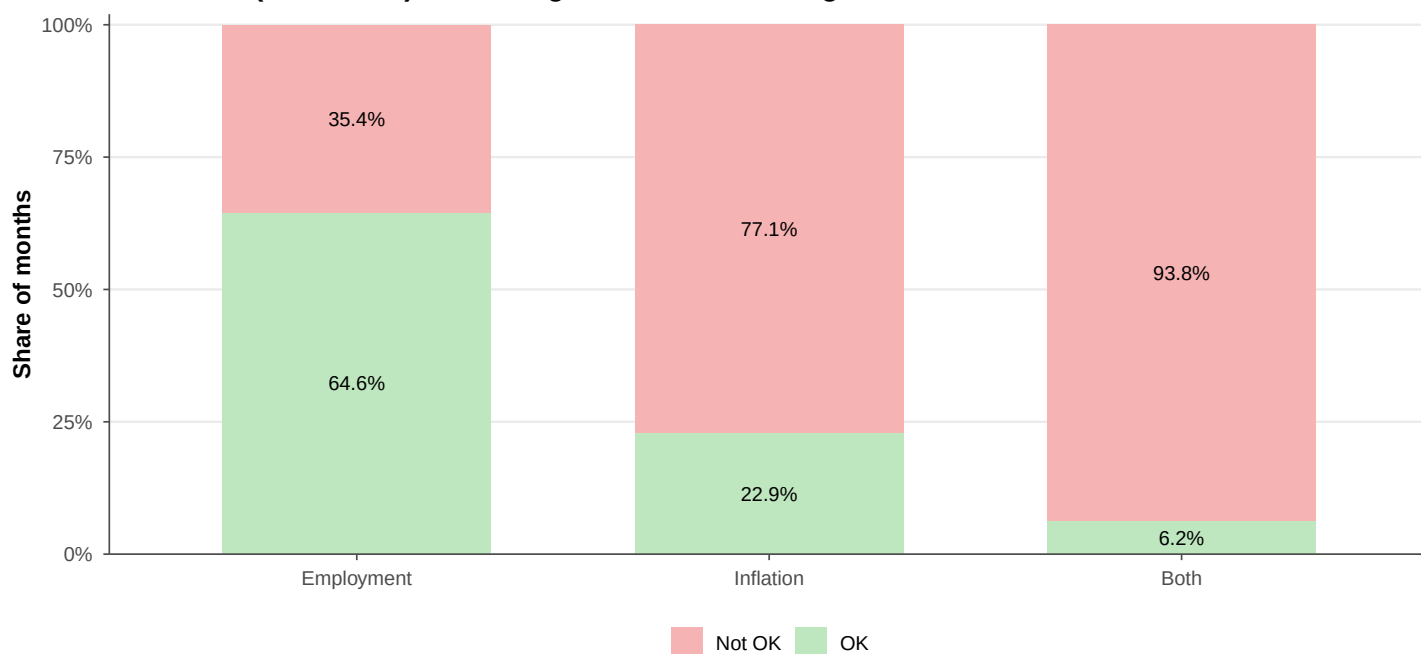
COVID era (2020–2023): Fed Funds vs Unemployment



COVID era (2020–2023): Fed Funds vs Inflation (CPI YoY)



COVID era (2020–2023): Percentage of months meeting benchmarks



11 Mandate Scorecard (Quantitative Answer)

We use a simple benchmark definition for “meeting both objectives”:

- Inflation near target: 1% to 3% CPI YoY (2% +/- 1pp band)
- Labor market strong: unemployment <= 5%

Table 1: 25-year mandate scorecard (monthly observations)

months_total	months_dual_ok	pct_dual_ok	months_infl_ok	pct_infl_ok	months_unemp_ok	pct_unemp_ok
299	83	27.8	178	59.5	147	49.2

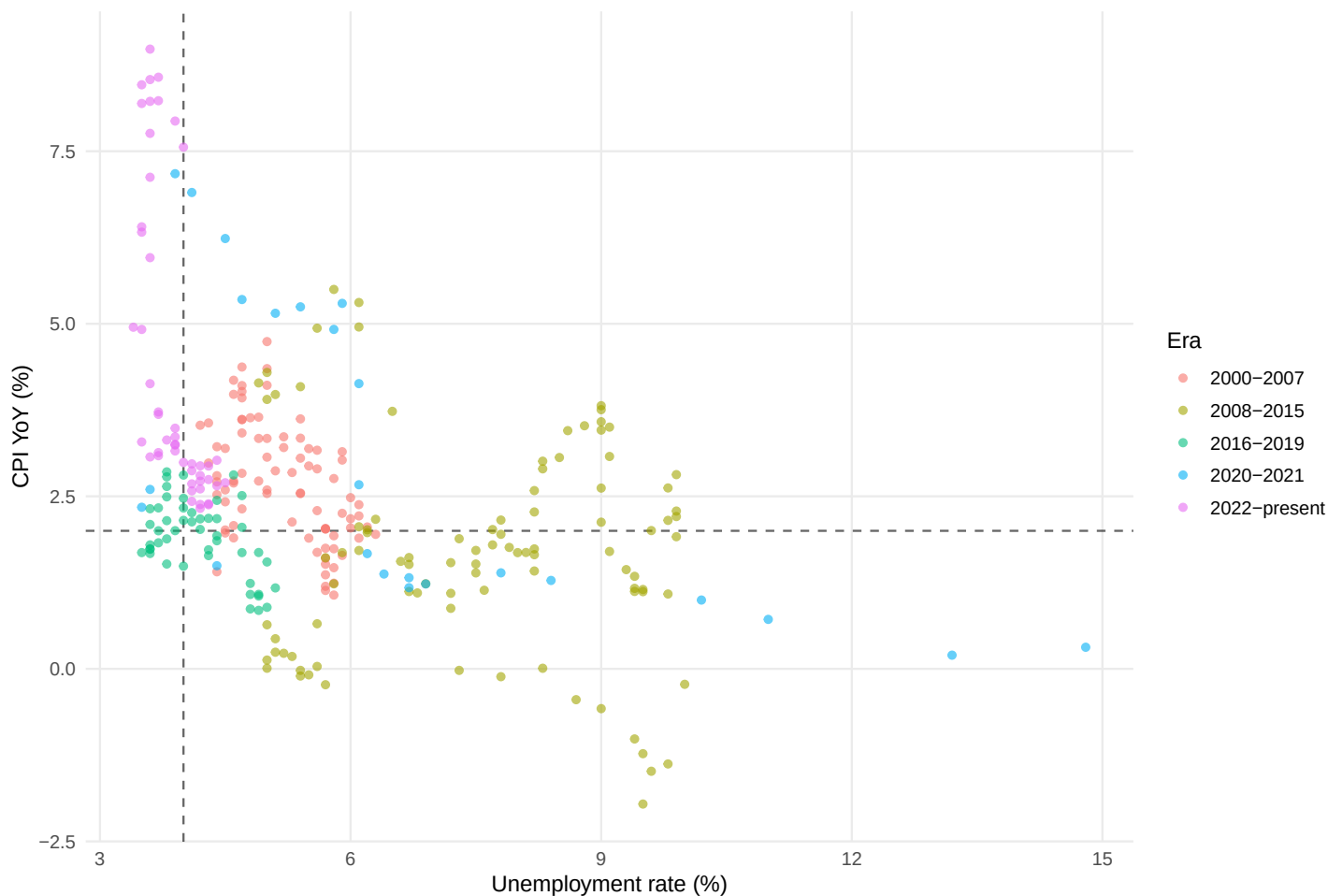
Table 2: Mandate scorecard by period (monthly observations)

period	months_total	months_dual_ok	pct_dual_ok	months_infl_ok	pct_infl_ok	months_unemp_ok	pct_unemp_ok
Dot-com bust (2000-2003)	35	4	11.4	29	82.9	8	22.9
Financial crisis + aftermath (2008-2012)	60	0	0.0	29	48.3	3	5.0
COVID era (2020-2023)	48	3	6.2	11	22.9	31	64.6

12 Visualization 2 — Tradeoff Map (Inflation vs Unemployment)

This “quadrant” view highlights when the U.S. was closest to the dual mandate (bottom-left).

Inflation vs unemployment: when did we get the 'good quadrant'?



13 Conclusion: Has the Fed Fulfilled the Mandate?

Short answer: Partially, but not consistently.

- The timeline and scatter show extended periods where one objective was met while the other was missed.
- The strongest dual-mandate performance appears in the late-2010s expansion.
- Major misses occurred in the Great Recession and pandemic labor shock (employment miss) and in 2021-2023 (inflation miss).

Overall, over the full 25-year window, the Fed appears able to **restore** the mandate after shocks, but not to keep both goals satisfied continuously.

14 Key Takeaways (Draft Slide Text)

Use these as the narration overlay for your Manim slides:

- **Best “dual mandate” stretch:** the late 2010s, when unemployment was low and inflation was not persistently elevated.

- **Employment misses:** recessions and shocks (2008–2010; 2020) when unemployment spikes regardless of inflation levels.
- **Inflation miss:** 2021–2023 surge (inflation far above a 2% benchmark while unemployment stayed low).
- **Timing matters:** the Fed reacts to inflation data; unemployment tends to move with a lag after costs/financing tighten.

15 Export Plots for Manim

16 Next Step: Manim Animation Plan (Outline)

1. Title slide: question + dual mandate definition
2. Timeline slide: fade in panels (inflation □ unemployment □ fed funds) + episode callouts
3. Tradeoff slide: animate pointer into “good quadrant” and compare eras
4. Lag slide: annotate right-lower quadrant as “hike □ future inflation falls”
5. Summary slide: mandate scorecard + caveat about lags/supply shocks